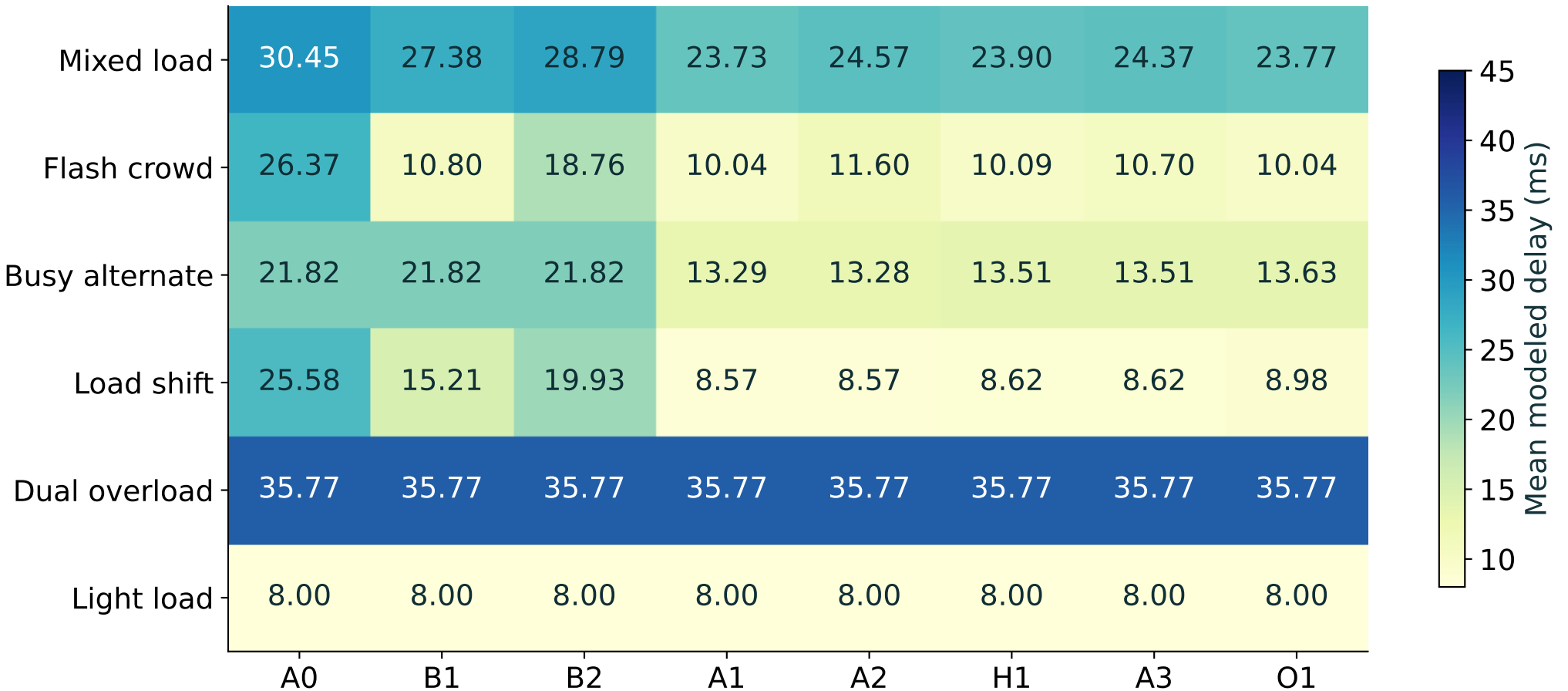
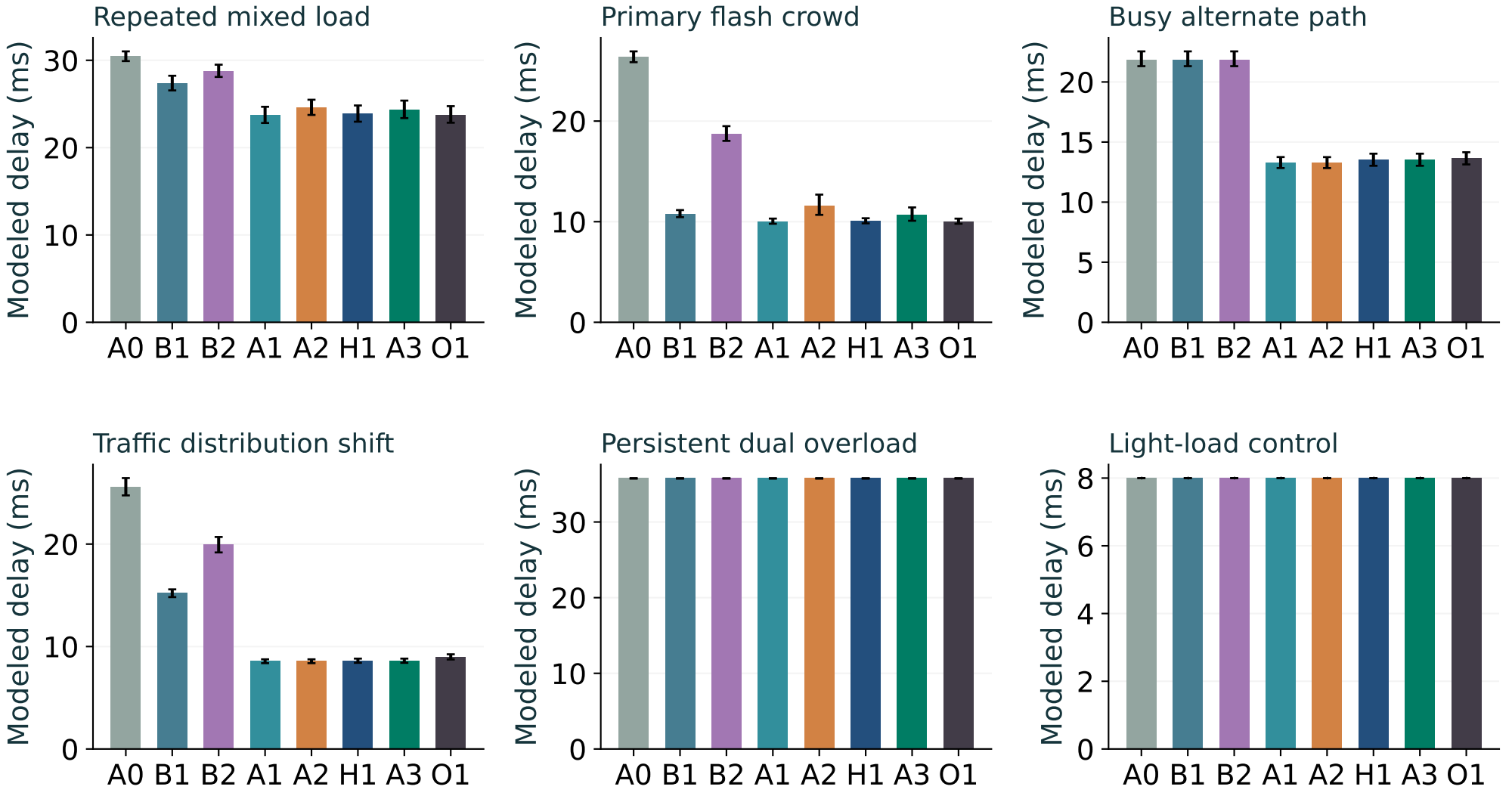
Delay across six traffic scenarios



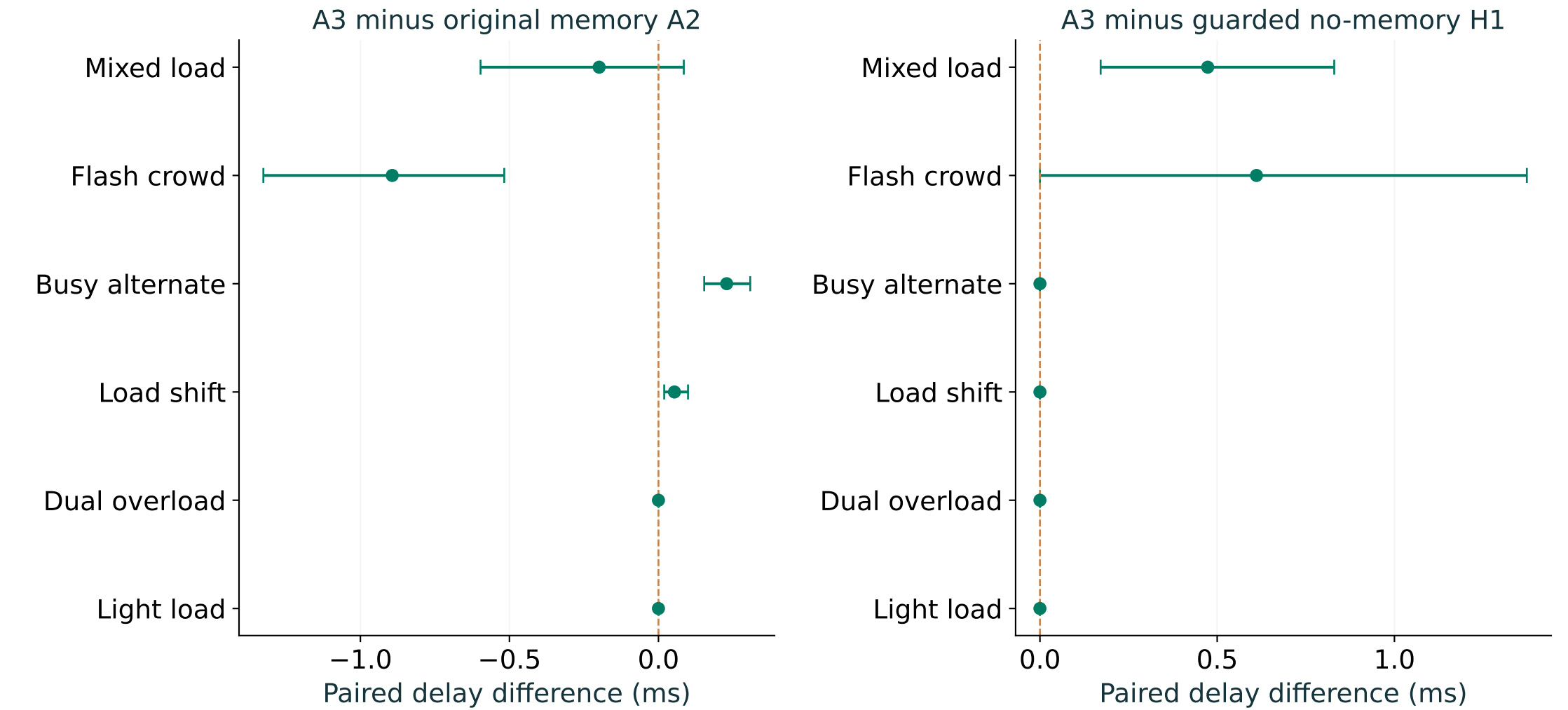
10 paired seeds × 120 windows × 8 controllers. Analytical simulation; lower delay is better. O1 is a myopic model reference.

Seed uncertainty by scenario



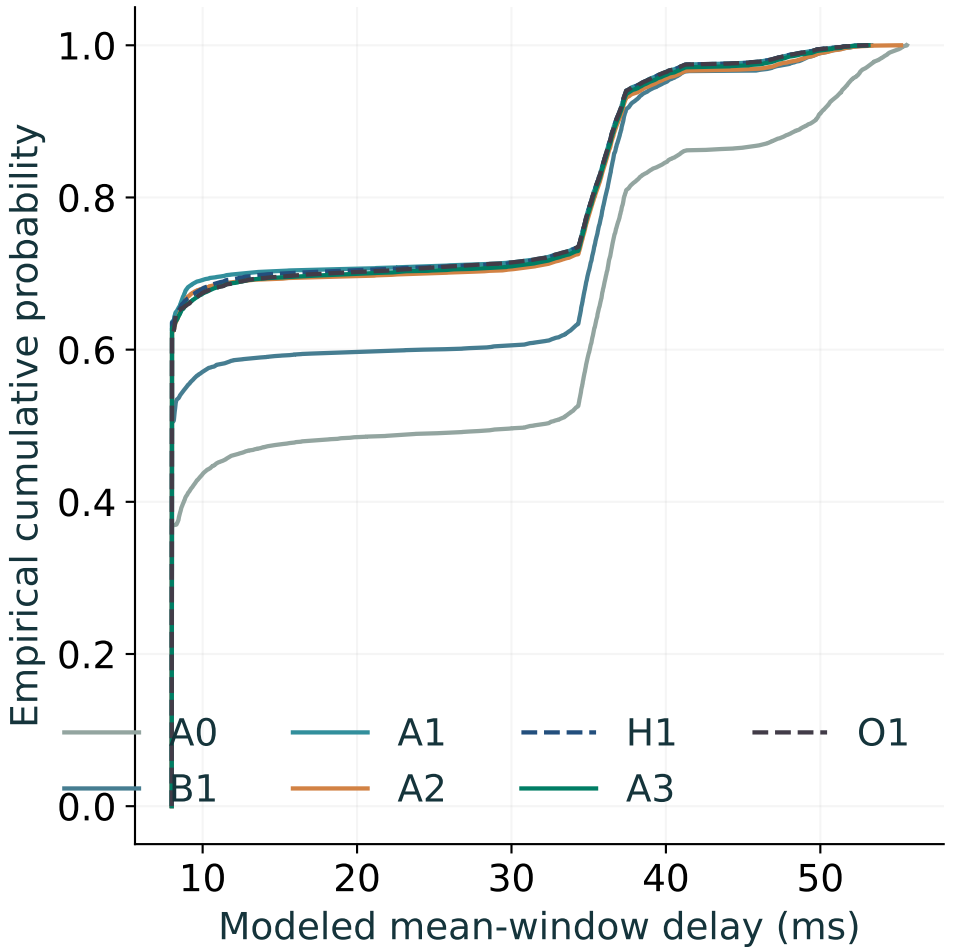
95% percentile bootstrap intervals across seed means (4,000 resamples). Exploratory comparisons; no multiplicity correction.

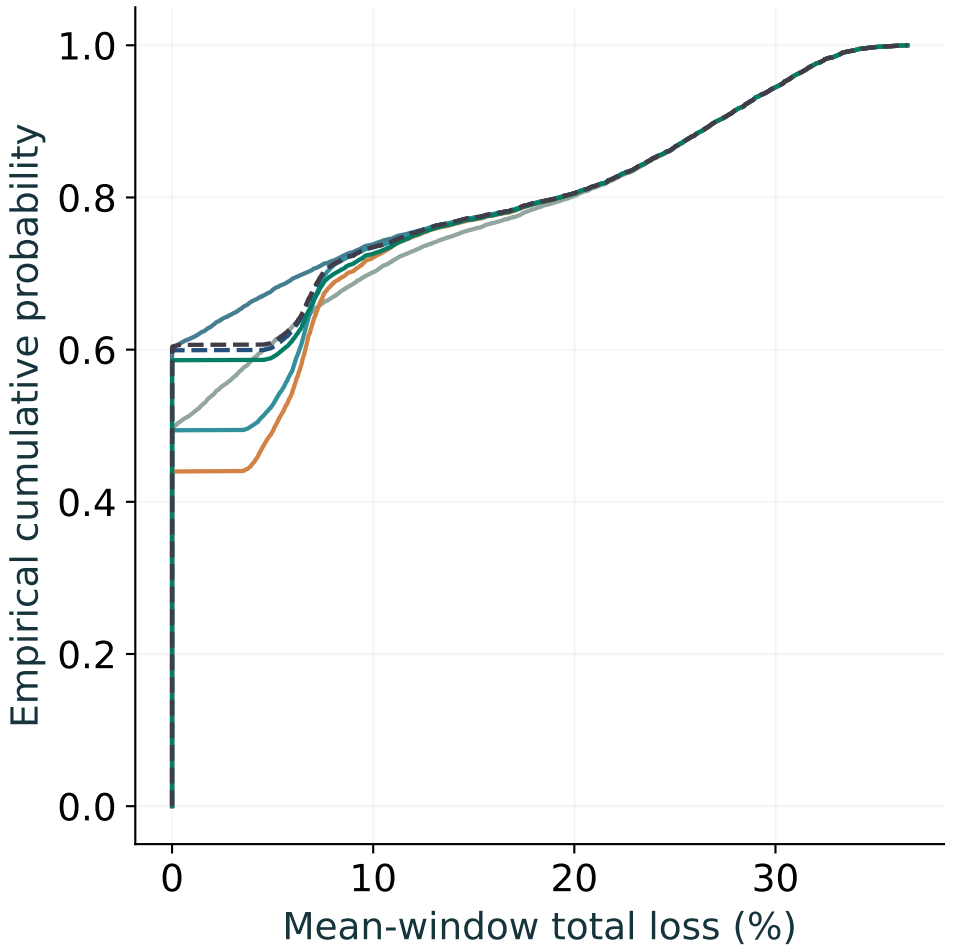
Memory effect and the guard ablation



Negative differences favor A3. Pairing uses identical scenario and seed. A3 versus H1 isolates memory after the common guard.

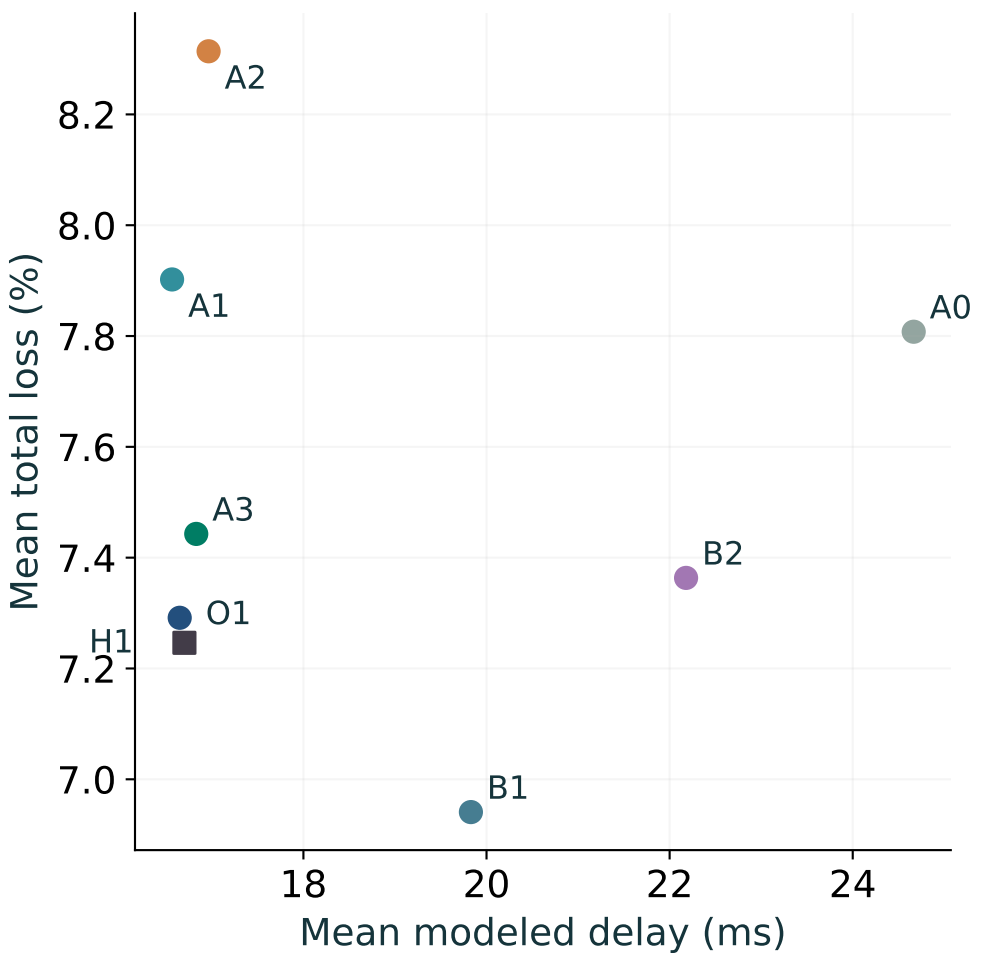
Distributions across all evaluated windows

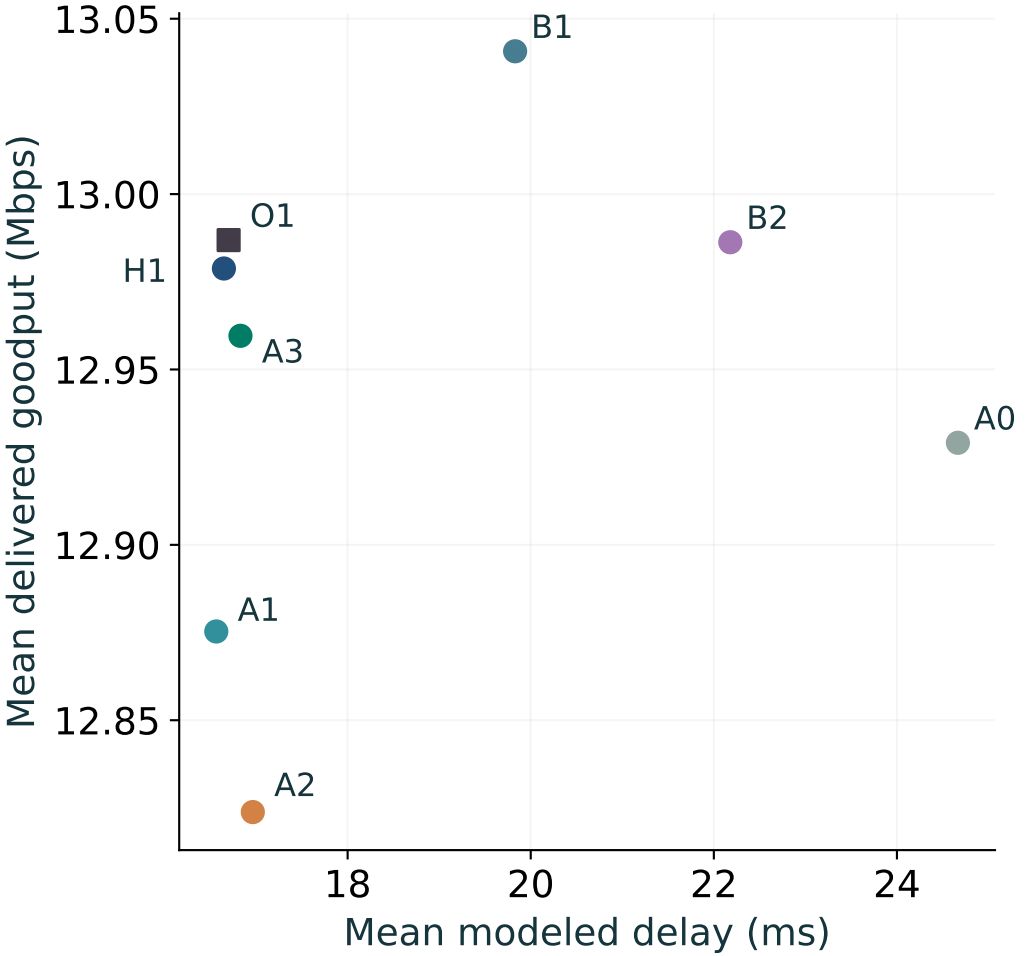




Equally sampled scenarios and seeds are pooled. These are distributions of model window means, not individual packet measurements.

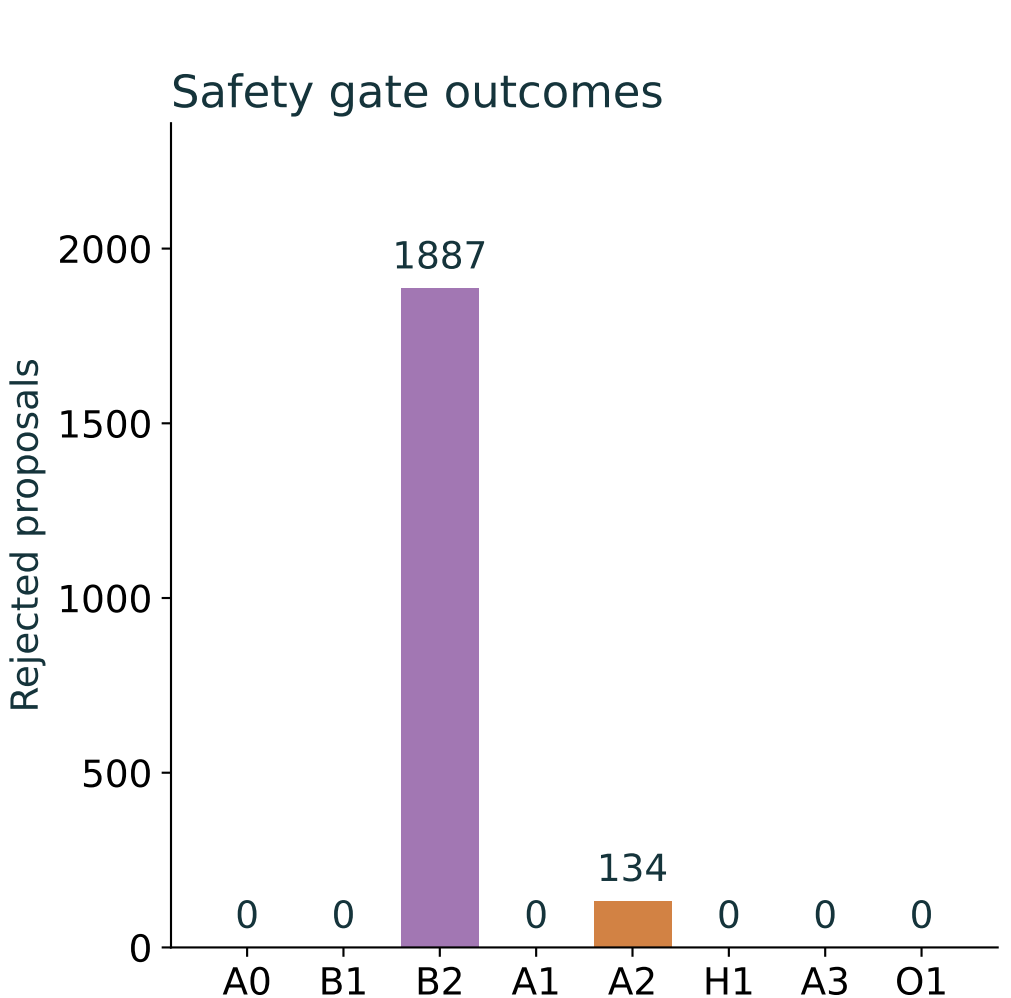
Delay must be assessed with loss and goodput





Each point averages equally over six scenarios and ten seeds. Prefer lower delay and loss, with higher delivered goodput.

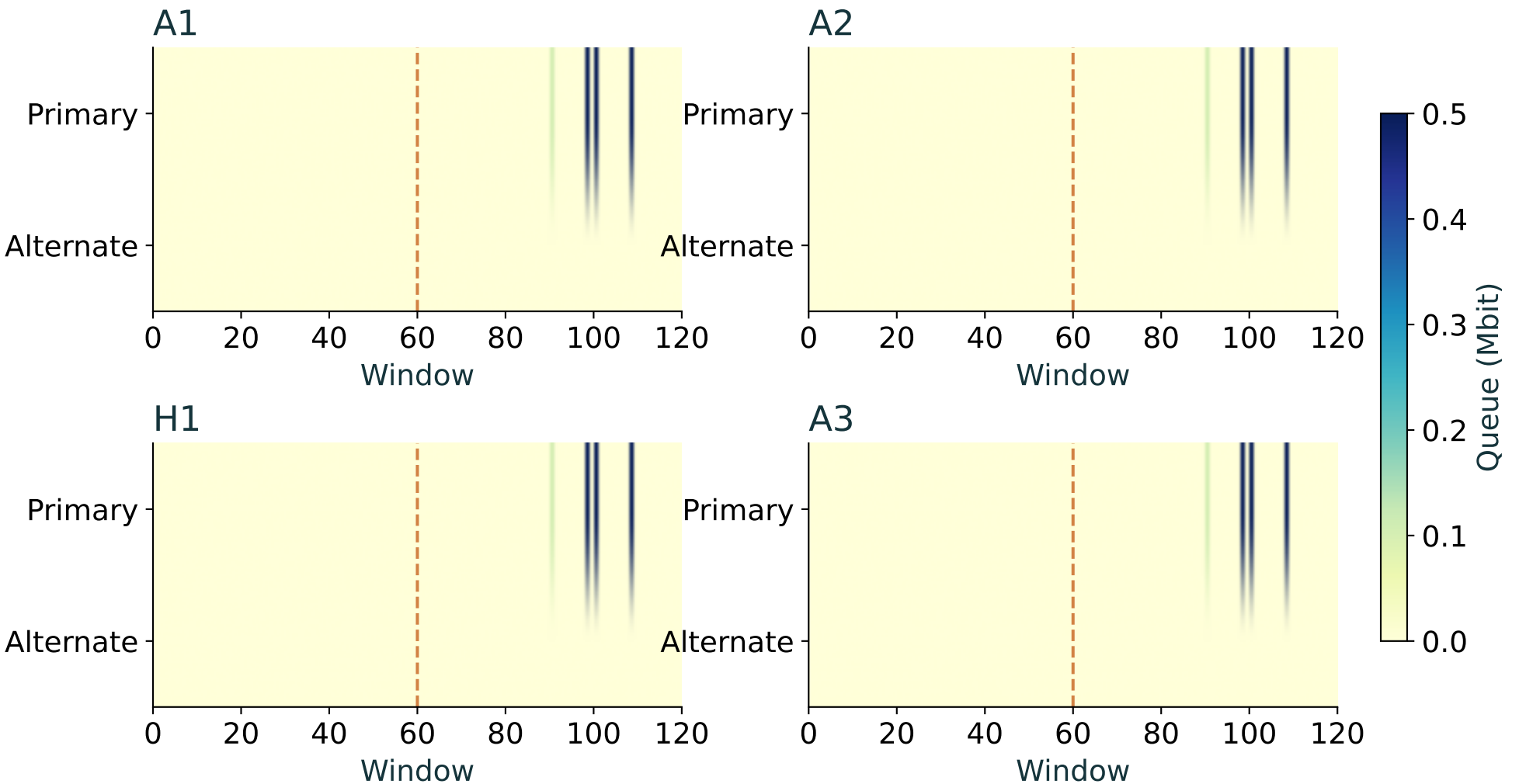
Guarded retrieval removes infeasible action reuse





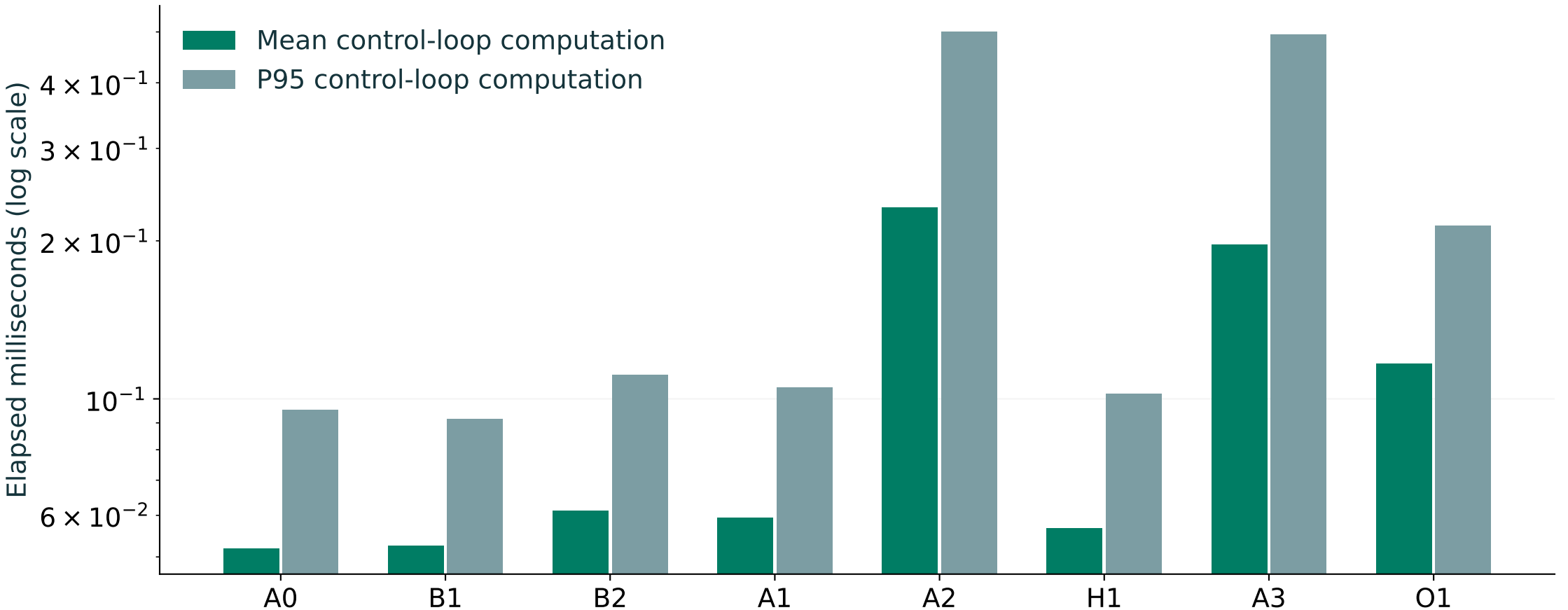
A3 validates retrieval candidates against current demand. B2 is an ECMP-style hash with the same independent safety gate.

Queue occupancy before and after a traffic shift



One illustrative trace (seed 1740); alternate background load increases halfway through. This figure is not an aggregate estimator.

Measured offline control overhead



Machine-dependent timing from .venv. Includes observation, decision, validation and measurement; excludes final batched memory persistence.